Plasma Spectroscopy

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Contents

Chapter 1

Lecture 1 - 01/10/2026

1.1 Course Introduction and Presentations

This is the first lecture of the course; we will proceed at a moderate pace, without overloading today's schedule. We will begin with mutual introductions and an introduction to the topics we will cover. My academic and professional background developed over seventeen years in the United Kingdom, after starting my activity in Catania between late 2024 and 2025. Throughout my career, I have held several seminars, including some in Glasgow and Belfast, focused on the applications of **spectroscopy** in different physical environments. My specific area of expertise concerns **plasma physics** and **plasma spectroscopy**, analyzing atomic energy levels in detail. In class, we have a heterogeneous audience. As happened last year, where we had students from different backgrounds (for example from New York) alongside astrophysics students and PhD candidates, I invite you right now to stop me and ask for clarification whenever a concept is not immediately understandable to you. During the lecture, we had the opportunity to listen to the students' presentations, including that of Karthik, a student from India, who shared his interest in 21 cm cosmology and **radio interferometry**, motivating his choice to study astrophysics with the desire to collaborate with the local research group.

1.1.1 Logistical Notes

Before delving into the study of energy, it is appropriate to define the breaks during the course hours: I prefer to take short breaks and resume quickly, but I leave the organizational choice to you. Regarding classroom management, a scheduling overlap emerged (in the 09:00 - 10:00 slot) with Professor Foumon's lecture. We will find a flexible logistical solution, possibly giving up my office if it is a remote lecture or reorganizing the spaces so as not to interrupt the teaching.

1.2 The *Solar Orbiter* Space Mission

Currently my research work is strongly linked to a large-scale space mission called **Solar Orbiter**, launched in February 2020. This mission is defined as

an *encounter mission*, as its objective is to cover the Earth-Sun distance and get extremely close to our star.

1.2.1 Onboard Instrumentation

On board the satellite, 10 scientific instruments are allocated, divided into two main categories:

- **4 *In Situ* Instruments**: their task is to collect and analyze local samples of particles, such as electrons and various ionic species.
- **6 *Remote Sensing* Instruments**: these consist of a set of advanced telescopes.

This is the first mission in which a group of telescopes is brought to such a close distance to the Sun. Such proximity entails enormous engineering challenges due to the extremely high temperatures (T) reached. For this reason, the probe, which is very massive, is equipped with a specific thermal protection technology, namely a **heat shield**, designed to protect the onboard electronics and, in particular, the delicate optics of the telescopes.

1.2.2 Orbital Profile and Global Observation

One of the most spectacular features of *Solar Orbiter* is its orbital profile. By exploiting complex gravity assist maneuvers with Earth and Venus, the probe is able to exit the ecliptic plane. Currently, the probe is performing a direct observation of the **solar poles**. This represents the best polar observation opportunity of the current century and is of fundamental importance not only for the study of solar physics, but also for the understanding of stars in general and for the dynamics affecting the Earth. Thanks to specific instruments, including spectrometers, *full-disk imagers* and the **Metis** coronagraph (dedicated to the exclusive observation of the solar corona), we are able to obtain complete coverage of our star: from the surface to the heliosphere and the interplanetary environment. This connects us to two macro-topics of modern physics:

1. **Cosmology**.
2. **Space Weather**: this field has immediate practical implications, as solar activity critically influences GPS systems, air traffic control, and power grids.

1.3 Course Prerequisites

To best tackle this applied spectroscopy course, the preparatory subjects closest to our topics are:

- **Structure of Matter**.
- **Quantum Mechanics**.

Knowledge of quantum mechanics is an essential requirement. During the course we will use the **Schrödinger equation** in all its possible formulations. While in Structure of Matter you have probably mostly analyzed the single-electron atom, in this course we will study the solutions for the single-electron atom, for the two-electron atom, and we will address **multi-electron** systems in detail.

1.4 Fields of Application

Although the course is oriented towards astrophysics, the methodology and spectroscopy we will study are of absolutely general validity and find application in countless high-tech environments:

- **Nuclear Fusion**: there is an inseparable link between controlled laboratory fusion and astrophysical fusion (which occurs naturally in the Sun and stars). There is a continuous and fruitful exchange of knowledge between these two sectors.
- **Laser Physics**: spectroscopic diagnostics are fundamental in high-power laser experiments (areas where, for example, laboratories in Canada have great expertise).
- **Particle Diagnostics**: applicable to the analysis of electrons, protons, and heavy ions.
- **Medical Applications**.

1.5 Practical Objectives: Plasma Diagnostics

Our course will be structured trying to answer nine fundamental questions or tasks. The first question, which I pose to you imagining that you are assigned a task as physicists, is the following: **How would you measure the temperature of a *solar flare*?** A *solar flare* is a gigantic and violent explosion originating on the surface of the Sun. During the lectures, we will learn to derive the thermodynamic and kinematic quantities from this type of events by exploiting spectroscopic data.

1.6 Solar Flares and Impact on the Terrestrial Environment

The solar flare does not manifest itself solely in the Earth's atmosphere; we can observe it very clearly from space in the ultraviolet (UV) range, especially by viewing the coronal loops through our filters. These events assume considerable importance because they emit charged particles that travel through interplanetary space. Such particles can reach the Earth and interact with the Earth's magnetic field: this phenomenon is at the origin of the Northern Lights, but can cause significant problems to terrestrial technological systems (such as power grids, communications, etc.). At this point, the previously discussed diagnostic problem arises: **How can we measure the temperature of a flare?** At first glance it might seem impossible. Someone might suggest sending a

probe directly inside the event; however, although probes get extremely close (up to extreme solar distances), the onboard instruments designed to collect direct samples (*in situ*) would inevitably disintegrate due to the intense heat.

1.6.1 Parallel with Nuclear Fusion Reactors

Having worked for two years in the field of nuclear fusion, I propose a parallel. Imagine you are in a controlled thermonuclear fusion plant: it is not physically possible to enter the reactor chamber, given the extreme radioactivity and the very high energy release. The environment is heavily shielded. Operations are conducted entirely from a remote *control room*, through which researchers "observe" the plasma confined inside the reactor using specific probes, cameras, and advanced diagnostic tools. This process of diagnostic abstraction is conceptually identical to what happens in the astrophysical field.

1.7 The Central Role of Radiation

Let's return to astrophysics and look beyond the Sun. How do we study objects located in deep space? There is only one medium at our disposal to study the entire Universe: **light** (or **electromagnetic radiation**). This is the only tangible element we possess. Even in the case of the Sun, despite the possibility of sending probes equipped with *in situ* detectors, almost all the essential information reaches us in the form of radiation. In this regard, I propose a sentence to you (although not strictly rigorous, it is very evocative): *"We listen to the voice of the plasma, which brings us information through light."* This sentence highlights an essential need: in order to interpret this "voice", it is necessary to have a tool capable of decoding the information carried by the radiation. And herein lies the peculiarity and power of **plasma spectroscopy**.

1.7.1 What Does Spectroscopy Measure?

Plasma spectroscopy is an extremely powerful investigation tool. Through the study of radiation from deep space (or from any other plasma source), it allows us to determine the physical and chemical state of the source itself. Specifically, by analyzing the radiation, it is possible to measure:

- The **temperature** (T).
- The **density** (ρ or n).
- The thermodynamic state (verifying whether the system is in equilibrium or in a dynamic state).
- The **magnetic field** (B).
- The **rotation** and internal motions of the source.
- The **chemical composition**.

Spectroscopic analysis acts as a true "footprint" of the observed source.

1.8 Course Structure and Physical Models

Let's move on to the formal objectives of the course, which define the goals we are going to pursue and the ways in which we will do so. These objectives are closely linked to the syllabus, which has been partially readapted. Originally, it was oriented towards the stellar field; however, the current approach is more generalist, so that it can be applied to all plasma contexts in the Universe, while maintaining the fundamental core intact (as can be seen by comparing the syllabuses of previous years).

1.8.1 Modeling and Interpretation of Radiation

The first objective is to provide future physicists with the mathematical and physical tools necessary to decode the radiation emitted by plasmas. To achieve this goal, we must build **physical models** of the studied environment to describe and reproduce the expected radiative emission. This step is critical because what we observe from space rarely comes from a single isolated source. I ask a question: when we observe a distant star from Earth, are we exclusively receiving the light emitted by that star? The answer is **no**. We simultaneously detect everything that stands along our **line of sight**. We see the starlight added to and altered by the emissions, and absorptions, of the interposed matter. This implies that, after defining the mathematical and physical framework of the model, we must understand how **atomic physics** processes generate the radiation (through atoms, ions, and molecules) along the entire path.

1.8.2 From the Model to Observations

Once the models have been developed, the next phase is the comparison with observational data to validate the model. However, this procedure proves to be complex: describing an atom with precision, even for simple cases (think of the increasing complexity passing from the one-electron atom to the multi-electron systems studied in Structure of Matter), is an arduous undertaking. Depending on the research sector, the modeling approach can vary:

- **In cosmology and stellar astrophysics:** the conventional approach is deductive, starting from the development of models to try to derive the expected observations.
- **In solar physics:** the reverse path (phenomenological method) is often followed. We start from the observation, build a basic model, and then progressively refine it to fit the data, asking at each iteration: "*What must we vary to reproduce what we observe?*"

1.9 Review of Plasma Physics

Before delving into the details, it is useful to recall what a plasma actually is. As is known, it is defined as the **fourth state of matter** and often roughly assimilated to a *hot gas*. This simplified definition is useful when setting up a fluid dynamics model, but it is not physically accurate. A plasma differs from a neutral gas by the following characteristics:

- It is a **macroscopically neutral** state.
- However, unlike an ordinary gas, **it is not composed exclusively of neutral particles**. A plasma is a fluid composed of ions (i.e., atoms stripped of one or more electrons), free electrons, and possibly protons, mixed with a residue of neutral particles.

The fundamental peculiarity of plasma is precisely the generalized presence of particles carrying an electrical charge, which makes it a state of matter impossible to maintain under standard environmental conditions on Earth (an environment saturated with free charges would be fatal to us). Plasmas exist in isolated conditions on Earth (such as in the luminescent gases inside fluorescent tubes for lighting, where the discharge is induced by the application of a potential difference). In such environments, although the total sum of positive and negative charges is macroscopically balanced, the charged particles move freely within the volume without undergoing an immediate and complete process of atomic recombination.

1.10 Plasma: The Fourth State of Matter

For this reason, plasma is defined as the **fourth state of matter**. It is a state that we rarely observe in our daily environment on Earth and that, to us, may appear unusual. However, it is the predominant state if we look outside our planet:

- It makes up the Sun and all the stars.
- It is present in vast quantities in deep space.
- It is obviously the fundamental environment in **nuclear fusion** experiments.

Precisely regarding fusion, we know that our ultimate goal (the "Holy Grail" of energy research) is to be able to build an efficient commercial fusion reactor, a technology that we currently do not yet possess. To achieve this, the study and management of plasmas are essential prerequisites.

1.10.1 Collective Electromagnetic Behavior

Secondly, there is a critical aspect that I believe you have already discussed in basic courses: charged particles within a plasma exhibit collective behavior. What is the main peculiarity of this state? Since plasma consists of charged particles, it **interacts directly and extremely sensitively with electric and magnetic fields**.

If an electromagnetic field is applied to an ordinary solid or liquid, the direct interaction is negligible or absent, unless under very specific conditions. For plasma, on the contrary, this interaction is constant and fundamental, precisely due to the presence of free charges.

Plasma does not react as a set of single, independent particles, but responds to fields (for example a radiation field) through a **collective behavior** of the entire fluid. **Practical example:** Let's consider a protostar (a forming star). The

inner core produces radiation that must pass through the surrounding clouds of matter. If these clouds are made of plasma (therefore of charged particles), the original radiation will interact collectively with the cloud, being altered. What we will detect will be different radiation, the result of the interaction between the protostellar emission and the interposed plasma.

1.10.2 Quasi-Neutrality

As mentioned previously, it is correct to state that plasma contains electrical charges, but it is equally true that the overall net charge of the volume is neutral. In experimental reality, the net charge is never strictly and mathematically identical to zero. As you may have noticed in teaching laboratories (for example in Laboratory 1 or 2), the results of experiments rarely match exactly with theoretical predictions. In these contexts, **approximations** come into play. If an approximation is physically reasonable and allows us to adequately describe the phenomenon of interest, then it is valid. For plasma, this very principle applies: the net charge can be approximated to zero. This condition is called **quasi-neutrality** and is one of the key concepts we will address today.

1.11 Course Program and Methodology (Syllabus)

Let us now move on to describe *how* we intend to achieve our educational objectives. I will show you how the various topics fit into the course syllabus. Despite starting from different points of view, the goal is to standardize knowledge and allow you to identify the fundamental processes. (The syllabus can also be consulted online).

The path is structured by addressing the following questions:

1. How to study radiation emission.
2. Identifying the countless amount of atomic processes operating in the plasma.
3. Analyzing fluid dynamics to understand whether the plasma is in equilibrium or not.
4. Building models (even very sophisticated ones) to answer the question: "How does plasma behave under different physical conditions?".

Points 1, 2, and 3 of the Syllabus focus on the **construction of the atomic and molecular model**. We must start from the foundations: the atom and the molecule. I decided to include some notions on molecules since they are of particular interest to those among you who are passionate about cosmology. In deep space, temperatures drop drastically. In these cold environments, the formation of molecules becomes possible; therefore, we will provide at least the basic concepts of molecular spectroscopy.

1.11.1 Atomic and Quantum Physics for the Course

To describe the atom, we need to employ quantum mechanics. We will study:

- **Energy levels** (E).

- **Wave functions** (ψ).
- The Schrödinger equation.
- **Spin**.
- **Coupling mechanisms**.

Coupling mechanisms (a cross between structural chemistry and physics) allow us to combine the angular momenta of orbitals to solve the atomic model. We will discuss **Russell-Saunders coupling (LS coupling)**, **spin-orbit interaction** and, albeit marginally, jj **coupling**. This study will initially be applied to the hydrogen-like (single-electron) atom. Since you have confirmed that you have already thoroughly covered the single-electron system in the previous course, I will dedicate a quick review to this topic, proceeding swiftly so that I can focus on **multi-electron systems**. Subsequently, we will extend these principles starting from the thermodynamic limit of approximation.

These first three points of the syllabus define the **fundamental quantities** necessary for the subsequent models.

1.11.2 Population Dynamics and Radiative Transfer

- The next point concerns the **population dynamics of energy levels**. Once the rates of all possible reactions are defined, we will develop a model describing how particles are distributed among the various energy states based on macroscopic characteristics (e.g. cold or hot plasma, *thin*optically thin plasma - or - *thick*optically thick).

Point 8 of the Syllabus, probably the most fascinating, concerns the **Radiative Transfer Equation**. This topic is of capital importance because it allows us to calculate exactly how radiation propagates from inside the plasma to the observer. The transfer equation is a flexible tool into which various plasma models can be inserted and then solved analytically or numerically. This entire conceptual path converges towards the primary objective: **understanding the meaning of a spectral line**. We want to be able to observe an emission line, study its width or intensity, and use it as a "window" to extract information about the source.

1.12 Advanced Modeling and Computational Calculation

The second general objective of the course (which covers **Points 4 and 5 of the Syllabus**) is to establish the operational connection between the theoretical model and the observable quantities. Models are often formidably complex. We need to define synthetic physical quantities that allow us, on the one hand, to simplify the model for the purpose of physical understanding and, on the other, to make it computationally tractable.

I assure you that the calculations required in this sector cannot be performed by hand; everything is delegated to the use of software run by specialists, given the enormous amount of parameters involved.

1.12.1 Practical Example: Calculation of Cross Sections

I bring my personal experience as an example. I dealt with the calculation of huge databases of **cross sections** (σ) for electron impact excitation processes. Imagine an electron striking an ion causing its excitation to a higher level: the probability of this happening is defined by a specific cross section.

This calculation must be reiterated for:

- Every element of the periodic table.
- Every possible ionization state of that element.

I worked specifically on **heavy elements**, from iron upwards. The computational load was such as to require the massive use of parallel computing architectures. The code was parallelized to make numerous computers work in synergy. A whole week of uninterrupted processing was often needed just to generate a single valid data file. A slight error meant having to start the calculation week all over again. This is to give you an idea of how quantum spectroscopic calculation requires a massive expenditure of machine resources (time machine).

1.12.2 Astrophysical Note: Iron Nucleosynthesis

I pose another astrophysical question related to these heavy elements: in the context of solar or space physics, what is the heaviest element whose spectrum we can commonly observe? Even if we could identify others in trace amounts, the primary element in these diagnostic observations is **Iron** (Fe). The iron atom (atomic number $Z = 26$) constitutes an evolutionary "bottleneck" for stars.

As you will have studied in the stellar evolution module, inside *high-mass stars*, thermonuclear fusion reactions (stellar nucleosynthesis) proceed by creating progressively heavier elements in the various *shells* of the core. This chain of exothermic reactions, however, inexorably stops at the synthesis of **Iron**. The reactions to fuse elements heavier than iron are in fact endothermic (they require energy instead of producing it). Consequently, iron accumulates at the center and is considered the heaviest stable element that we can widely observe in ordinary stellar atmospheres to perform large-scale diagnostics.

Through **Points 4 and 5**, therefore, we are going to quantify the rate coefficients and the transition probabilities necessary to analytically and computationally manage this type of chemical-physical environments.

1.13 Quantification of Spectroscopic Coefficients

We will now examine the oscillator strength. You are probably already aware of it: it represents the oscillation of the atom coupled with the transition strength. We will see multiplets, cross-sections, and collision rates. We will study excitation, de-excitation, ionization, and recombination.

If we have enough time, we will address another extremely important process: **photoionization** and **charge exchange**.

Why did I explicitly include charge exchange? Usually, not much charge exchange is observed as a dominant transition; however, in which specific space

environments would you expect to observe this process? By "charge exchange" we mean a process in which two atoms or ions collide and exchange charge (for example, an ion transfers an electron to a neutral species or to another ion). There is an emblematic case. Obviously, the process occurs everywhere, from stars to galaxies to the interstellar medium, including nebular media (nebulae). But there is a particular object in our Solar System where charge exchange has been clearly observed and is fundamental: **comets**. The observation of the spectrum of comets is dominated by the signatures of this phenomenon. Therefore, should any of you be interested in the physics of minor bodies, you will find this concept useful.

1.14 The Line Profile

Another topic that we have previously mentioned but will discuss in detail is the **line profile**, covered in point 7 of the syllabus.

When we observe a spectrum, what is its graph materially? I remember that during my university days, professors constantly repeated this definition: *A spectrum is the intensity as a function of frequency or wavelength.* This is the basic concept. We will therefore have:

$$I(\lambda) \quad \text{or} \quad I(\nu) \tag{1.1}$$

where I is the intensity and the independent variable can be the wavelength (λ, measured for example in Ångström Å), or the frequency (ν or sometimes indicated with f). The two representations are obviously interchangeable by proceeding in opposite directions, given that $\lambda \propto 1/\nu$.

Suppose we plot a graph of $I(\lambda)$. Ideally, if the radiation comes from a quantum transition between two discrete energy levels of the atom, the line should appear as an infinitesimal impulse (a Dirac delta function): a single, extremely narrow vertical peak. In observational reality, however, the lines present themselves with a broadened shape, called a **profile**. This profile can be described mathematically, for example, using a Gaussian, a Lorentzian, or a Voigt function (Voigt profile), and can present itself narrower or broader depending on the physical conditions.

1.14.1 What Determines the Line Profile?

The line profile is crucial because it conveys a wealth of information superior to mere intensity:

1. **Instrumental Characteristics**: In the first instance, the profile is limited by hardware resolution (instrumental profile). No instrument is perfect; the spectral resolution tells us how precisely the instrument can separate nearby wavelengths.

2. **Plasma Dynamics and Thermodynamics**: Once the instrumental profile is known, we can subtract it (technically, perform a *deconvolution*) to obtain the intrinsic profile of the line emitted by the object. This intrinsic profile reveals the kinetics of the emitting plasma:

    - Macroscopic motions and turbulence.

- Temperature (via thermal Doppler broadening).
- Pressure.
- Presence of electromagnetic fields. For example, the magnetic field induces a **Zeeman effect** (Zeeman splitting), which divides the line into multiple components or drastically modifies its overall profile.

As can be guessed, the very presence of a specific line informs us about *which* chemical elements make up the plasma; however, studying its strength and broadening allows us to conduct a complete physical analysis of the environment.

1.15 Diagnostic Applications and Quantitative Tools

The final part of the course (Point 9 of the Syllabus) is dedicated to **practical applications**. This is the section that will be most useful in your future career (master's thesis, doctorate, or work in astrophysics): acquiring data, optimizing it, and understanding its physical meaning.

Having built the theoretical model and obtained the observations, we unite the two worlds by extracting the diagnostic parameters. We will focus only on a few fundamental techniques. Two main examples are the derivation of the electron density (N_e) and the temperature (T) by exploiting the **spectral line intensity ratio** (line ratio).

1.15.1 Density Diagnostics: The Case of Oxygen V (OV)

Take for example an analysis conducted with the **SPICE** spectrometer on board Solar Orbiter. One of the observed ions is **Oxygen V** (OV, that is, 4-times ionized oxygen, according to spectroscopic notation). I will dedicate a more technical discussion to this ion in the next lectures. In the OV spectrum, a **multiplet** composed of various lines is observed. Among these, there are two specific lines whose intensity ratio is *insensitive* to temperature variations, but is **strongly dependent on density variations** of the emitting plasma.

By measuring the intensity of these two lines (integrating the area subtended by their line profile) and calculating their ratio R, we can derive the density. On the theoretical side, models are solved for a wide range of densities to produce a theoretical curve.

If we plot on a graph:

- The x-axis (abscissa): the **electron density** N_e (expressed in cm^{-3}).
- The y-axis (ordinate): the **intensity ratio** $R = I_1/I_2$.

We will obtain a curve $R(N_e)$. By placing the numerical value of the measured ratio R_{obs} on this curve (calculating the relative statistical error and propagating the uncertainty), we can extrapolate the corresponding electron density value of the target plasma.

This procedure, although schematized, constitutes an invaluable investigation tool. The same logic applies to determining the temperature, carefully selecting lines whose ratio is sensitive to T but independent of N_e. We will study the principles of atomic physics that justify the different behavior of these lines.

1.15.2 Chemical Abundances and Space Weather

A further applicative step (if time permits) will be the calculation of **chemical abundances**. Abundance analysis is complex, but in our solar research group it is widely practiced as it represents the only reliable trace to connect remote sensing (spectroscopic) measurements with the detections of *in situ* instruments. If, for example, we observe a mass ejection (CME, Coronal Mass Ejection) and remotely determine its isotopic or elemental abundance, and we later manage to measure the same abundance pattern via the instruments onboard the satellite when the plasma hits it, we can ascertain without a doubt the origin of that plasma. This allows us to predict the dynamics of the interplanetary plasma with several days of advance compared to the impact with the Earth's magnetosphere, providing a vital *forecast* service for Space Weather (for the safeguarding of GPS, power grids, and air traffic control).

1.16 Information About the Exam

Finally, I would like to give you some operational notes regarding the material and the exam. I noticed the presence of a bug in the online Syllabus system: selecting the English language, the page redirects to the top and does not properly load the extended view of the program. I assume it is an issue currently being resolved by the University. In any case, the full text is viewable, so you can always use it as a reference and ask me questions.

Regarding the **final assessment** modality, I reflected on it yesterday and am still defining the optimal details. The previous professor set the exam solely in the form of a one-hour oral exam, proposing three main questions and two secondary questions. Personally, I believe this formula is not the most effective one, so I will adopt a different verification methodology that I will communicate to you shortly.

1.17 Exam Modality and Course Structure

Having outlined the conceptual complexity of the subject, I consider it fundamental to recalibrate the final assessment modality. Since this course has a strong theoretical foundation – albeit preparatory to applied diagnostics – I understand that the systematic study of the entire volume of topics would be excessively burdensome if confined to a single intensive final test.

The previous approach, based exclusively on the oral exam in a single session, does not seem to me the optimal didactic solution. My proposal is the following:

1. **Lightened Final Oral Exam**: The final exam will take place regularly in the form of an oral interview (within which I will include the usual three main questions). However, it will not cover the entire syllabus, but will focus on the final modules (interpretation of line profiles and diagnostics), which require a more mature and discursive approach.

2. **Intermediate Tests (In itinere)**: During the semester I will introduce short *in itinere* tests. These tests will have a practical slant and will be based on solving exercises (for example: determining the spectral term, calculating the coupling for a given energy level, etc.).

I believe that these calculations, which are fundamental for assimilating the structural and quantum physics foundations of the course, are much easier (and more rewarding) to solve on paper with adequate time, rather than being verbally presented during a face-to-face interrogation with the professor. By tackling and passing these short assignments, the relevant part of the syllabus (the first sections of the Syllabus) will no longer be subject to verification at the final exam.

I know well how the preparation of vast exams such as Mathematical Physics or Quantum Mechanics can require entire semesters of study (I still remember my written Mathematical Physics exam lasting four hours for four complex exercises). By adopting this division, I hope to help you assimilate the material progressively and present yourselves ready for the summer session (July) exam dates. This is my didactic intention. However, I invite you to provide me with *feedback* (also in the upcoming lectures); if you consider this method not very functional, I am fully available to re-evaluate the exam structure based on your needs.

1.18 Real Applications of Spectroscopy: Beyond the Classroom

To conclude this introductory lecture, I wish to show you the real potential of what we are about to study, so that it is clear to you that the theoretical apparatus is not an end in itself, but represents an essential tool for physics and high-tech industry.

To this end, I will present the key concepts extracted from one of my academic seminars entitled: "*From Astrophysics to Tokamak Reactors: Plasma Diagnostics Applications*". I do not expect you to immediately grasp all the technical details – the presented models are very complex – but I ask you to focus on the common thread: **the transversality of the technique**. The spectroscopic methodologies born to decipher the Universe are the same ones employed today for the control of nuclear fusion on Earth.

In this seminar, the contributions of **Prof. Hugh Summers** (my PhD supervisor in Scotland and a brilliant mind in the field) and **G. Doyle**, operational expert in Armagh (Northern Ireland) in the field of diagnostics, are highlighted. The methodologies we will see have been widely used at **JET** (Joint European Torus), the largest European fusion experiment located in Culham, in the United Kingdom, now close to closing to make room for the new generation. The heir to JET is **ITER**, a gigantic experimental reactor currently under construction in the south of France (near Marseille). ITER represents the crucial step towards the development of future industrial fusion reactors (the "DEMO reactors"), which will perhaps see the light towards the end of our careers.

1.18.1 Advantages of Radiative Diagnostics

In both astrophysical and laboratory settings, the foundational concept is the same: to derive the physical and mechanical parameters from the **emission (or absorption) profile** of the radiation. (As is known, just as emission spectra exist, absorption spectra exist, which follow the same physical logic).

Why is radiative spectroscopy the diagnostic method of choice for plasmas? Because it is a **non-invasive** investigation method. When you introduce physical probes (such as Langmuir probes) into a laboratory plasma to measure its potential, you inevitably perturb the plasma itself, modifying the local electromagnetic equilibrium (as established by the cardinal principle of measurement discussed in Experimental Physics). Furthermore, probes return purely local (point-like) measurements, requiring probe arrays to map a volume, which is often impossible in ultra-high temperature reactors. Passive spectroscopic observation (tele-detection or *remote sensing*) completely eliminates the problem of perturbation and allows extended volumes to be analyzed.

1.18.2 From Astrophysics to the Laboratory: The Sun in the Extreme Ultraviolet (EUV)

Let's start with the astrophysical object par excellence: the **Sun**. The standard image we have of the Sun (in the visible) is that of a quiet, yellow disk. However, if we observe the Sun through space filters operating in the **Extreme Ultraviolet (EUV)**, the appearance changes radically. Observing the images recorded in the EUV range, we no longer see a uniform surface, but a chaotic, dynamic, and violent three-dimensional structure, dominated by **coronal loops** and prominences. All this dynamics, which is almost completely hidden in the visible, is governed by solar **magnetic fields**.

This complex interaction between plasma and magnetic fields – the origin of flares and high-energy radiative emission – will be the subject of your next Solar Physics course. For our part, we will learn to read the information encoded in the emitted radiation to fully understand the extreme environment from which it originated.

1.19 Solar Dynamics and Space Weather

These images derive from data collected by the **SDO** (*Solar Dynamics Observatory*) space satellite. Among the various instruments on board, there is the **AIA** (*Atmospheric Imaging Assembly*). This instrument is an *imager*: it acquires images of the entire solar disk (full sun) allowing the extraction of details at extremely high spatial resolution, operating on different bands of the Extreme Ultraviolet (EUV). The observed bands range from those corresponding to relatively "cold" plasmas to those of extremely hot plasmas.

For example, in the image you see colored red, we are observing the **Helium II** (He II, singly ionized helium) line at 304 Å (EUV). This line is indicative of the solar **chromosphere**. "Bubble" structures or "granules" can be clearly noticed. Those of you with notions of stellar physics will know that below the solar surface there is a convective zone (convection zone) where hot plasma rises and then sinks once cooled, just like a boiling liquid. These convective bubbles leave a clearly visible signature (imprint) in the overlying chromosphere, and that is exactly what the AIA reveals at this wavelength.

The image colored in yellow, observed previously, shows even more external layers. The Sun, in fact, is stratified:

1. **Photosphere**: the apparent surface, the one optically visible from Earth.

2. **Chromosphere**: the layer immediately above.
3. **Transition Region**: a very thin layer.
4. **Solar Corona**: the outermost atmosphere.

The dynamics in these regions govern the expulsion of matter into space. Strong magnetic explosions, called flares, can trigger phenomena known as **Coronal Mass Ejections** (CME). A CME projects enormous quantities of plasma (radiation and charged particles) into interplanetary space, generating the space weather that influences the terrestrial environment. Despite knowing the Sun well as our closest star, many unsolved mysteries remain regarding its functioning.

1.19.1 Thermal Stratification and Spectroscopic Diagnostics

The SDO satellite orbits at about 1 AU (Astronomical Unit) from Earth and provides continuous monitoring (24 hours a day) at very high cadence (high-cadence), which is fundamental for capturing the rapid dynamic evolution of events.

The various AIA filters allow us to perform a stratigraphic scan of the solar atmosphere. How is it possible to know which layer we are observing? Because **every spectral line has its emission peak at a very specific temperature**. It is not possible to observe the emission of a certain ion if the local plasma does not have the temperature necessary to produce and maintain that ionization state.

Let's analyze the thermal profiles:

- The photosphere emits like a **black body**. The continuous visible radiation of the Sun (the typical "yellow" color we see with the naked eye, although the peak is actually shifted more towards green-yellow) corresponds to a surface temperature of about 6000 K (often cited around 5800 K).
- The He II lines at 304 Å (chromosphere) reach their peak around 80000 K (often rounded to 10^5 K in the upper transition region). In these layers, the analysis of the lines allows us to map the plasma density as well. Moving up in altitude, the temperature increases while the density gradually decreases.
- The line at 171 Å is produced by Iron IX (Fe IX). This emission marks the boundary between the upper chromosphere, the transition region, and the lower corona.

Precisely in the **Transition Region**, a phenomenon occurs that still today constitutes one of the greatest enigmas of astrophysics (the coronal heating problem): in a very narrow atmospheric thickness, the plasma density drops drastically (from 10^{11} to 10^8 cm^{-3}), becoming very thin, while **the temperature soars** vertiginously from the photospheric 6000 K to over **a million degrees** ($> 10^6$ K) in the corona. Physicists believe that the energy necessary to sustain these temperatures comes from the dissipation of the strong

coronal magnetic fields, but the exact mechanism has not yet been completely reconstructed or modeled in all its details.

1.20 Nuclear Fusion and Magnetic Confinement

Let us now move from astrophysics to the laboratory environment. The most severe operational limit of a thermonuclear fusion reactor is duration. The plasma fuses but, due to fluid dynamic and thermal instabilities, the fusion conditions (in the past) could be maintained only for **microseconds**. An enormous power release that lasted a fraction of a second is unusable for an industrial plant (power plant) that needs to produce continuous electric current. You have probably happened to read recent press releases or social media news announcing "fusion duration records". We have gone from microseconds to milliseconds and, recently, we are beginning to achieve a few seconds of stable confinement. However, for a commercial reactor, the challenge of maintaining the plasma remains immense.

In the laboratory, we have the advantage of being able to alter the parameters of the model to directly verify our theories, something obviously impossible when observing a distant star, where diagnostics are purely observational.

There are different approaches to controlled fusion.

- **Magnetic Confinement**: This is the approach adopted by the **Tokamak**. This device (which has a characteristic toroidal "donut" geometry, although the internal physics is very complex) uses gigantic magnetic fields to "confine" and keep the plasma trapped and detached from the reactor walls. Major projects like JET and ITER use this architecture.
- **Inertial Confinement**: Although the civil nuclear power industry is stalled in Italy, scientific research is very active. For example, in Catania, research groups pertaining to nuclear physics (and covered in the Foundations of Nuclear Physics courses) also operate in an inertial confinement regime (typically obtained through the implosion of fuel capsules induced by ultra-high-power lasers).

New reactor models (including ITER) incorporate a vital module called the **Divertor**. Generally located at the bottom of the Tokamak, the divertor acts as an "exhaust system". The central plasma is terrifyingly hot (in the order of tens or hundreds of millions of degrees, even hotter than the inside of the Sun). When the particles (or the excess helium produced by the reaction) are diverted towards the divertor, they must obligatorily be cooled before they can be extracted, otherwise they would destroy the walls. This cooling is achieved by injecting puffs of heavy gases (such as neutral helium, argon, neon, etc.) that radiate the energy, lowering the local temperatures. Our spectroscopy is essential to monitor the efficiency of this process.

1.21 Atomic Physics and Ionization Diagrams

In all these contexts, **Atomic Physics** is the key to interpreting the spectrum and deducing the properties of the emitting source. I will show you a graph

that has considerable historical importance for me and for my department in the United Kingdom. The researcher who occupied my office before me was **Jim Lang**, an absolute pioneer in the development of atomic physics for plasma diagnostics. Lang worked for years on the **SOHO** (*Solar and Heliospheric Observatory*) satellite, launched in 1995 and still partially in operation, which was the first real space mission to make intensive use of this specific type of atomic models applied to astrophysics. Lang asked himself: "*How can we correlate the radiative emission of different ionic states both to solar events and to laboratory machines?*"

From these studies comes the graph reproduced here (diagram of wavelength versus formation temperature).

1.21.1 Analysis of the Thermal-Spectral Diagram

In the graph, the x-axis represents the wavelength (λ), ranging in the EUV (from 200 Å to 1600 Å), while the y-axis indicates the **Formation Temperature** of the ions. Every single "point" visible on the graph corresponds to a **specific ionization state** of a chemical element (e.g., iron, silicon, oxygen).

The usefulness of this plot is immediate: it traces the thermal stratification of the plasma based on the emitted radiation.

- **Cold Plasmas**: In the lower part of the graph (low temperatures, corresponding to the lower solar atmosphere) we find neutral species or those with a low degree of ionization: neutral atoms (hydrogen, helium), monovalent ions and, as already mentioned, even molecular traces in the coldest regimes. If your spectrum presents only these lines (or if the Silicon III line, Si III, to name one, dominates), it means you are observing a relatively cold environment.

- **Hot Plasmas**: Moving up along the temperature axis (passing the chromosphere and arriving in the corona), particles lose more and more electrons due to thermal collisions. We therefore begin to detect extremely high ionization states: iron, for example, will present lines corresponding to Fe XIII, Fe XIV or even Fe XX and higher.

Therefore, the simple presence or absence of specific spectral lines in the UV range provides, as a first approximation, the first fundamental diagnostic parameter: it defines, at a glance, the thermal order of magnitude of the plasma region under observation.

1.22 The Intersection between Theoretical Model and Observation

Everything we have illustrated so far constitutes the true connection between theory and experiment, and it is the goal we wish to achieve in the first part of this course. The purpose of our model (the "theory") is to describe the physical environment in order to interpret the "observations" (the measurements). For you physicists, the most fascinating field of application is likely deep space, but the same concepts apply to laboratory diagnostics.

These models must be constantly updated, as new spacecrafts and increasingly powerful new instruments are launched. You have certainly followed the progress of the **James Webb** Space Telescope (JWST). When the first images were published (which a colleague of mine also worked on), the enthusiasm of the scientific community was enormous. But the increase in instrumental performance entails a critical need: if we have excellent instrumentation, we must have **theoretical modeling of the highest quality**. Otherwise, we will observe new phenomena in the data, but we will not be able to understand their physical nature.

1.22.1 The Example of Spatial Resolution: Solar Orbiter

I bring you the example of Solar Orbiter. In the very first acquired images, we noticed some very small *brightenings* (micro-scale flares), never seen before. This was possible because Solar Orbiter reached an extraordinary spatial resolution: about 200 km on the Sun. In the past, instruments could resolve details on the order of 1700 km per single pixel (or image point).

To understand the magnitude of this technological leap, imagine terrestrial geography: the distance between Milan and Catania is about 1500 km. The distance between Messina and Catania is about 100 km. With the old resolution (one pixel = 1700 km), all of Italy from Milan to Catania collapsed into a single point; you could not distinguish anything inside it. With the new resolution (one pixel = 200 km), you can suddenly distinguish individual cities along the route. From the point of view of solar observation, this leap in resolution brings out unprecedented structures and dynamics (e.g., micro-flares) that the old models were not structured to explain. Therefore, the need to update the theoretical apparatus emerges.

1.23 Limits of Low-Density Models

The very first model we will deal with is the **Coronal Model**. It is a model developed for plasmas with a density close to zero (zero density model) and is widely used in astrophysics for deep space. Its advantage lies in its extreme simplicity: by including only a very few atomic processes, it works excellently for very thin plasmas.

However, as soon as we point the new high-resolution instruments towards denser solar regions (or if we apply the model to a fusion reactor), the Coronal Model fails miserably. It is imperative to develop new models that answer the question: *"What happens when density increases?"* Studying the density increase is vital not only for fusion, but also to understand **solar flares**. During a solar flare, the local density increases. We want to understand the evolution of the flare to model it and predict its effects; similarly, we want to understand the innermost and densest layers of the solar atmosphere.

1.24 Modern Approach: The Bundle Model

The fundamental progress in this area dates back to a 2006 publication, a method on which I later based my own doctorate. The central problem is this: in Structure of Matter you studied the model for a single isolated atom. But in

a real plasma we do not have a single atom: we have an Avogadro number of atoms, consisting of very many electrons, which collide and transition between countless energy levels.

How many energy levels does an atom possess? The theoretical answer is **infinite**. A rigorous computational model should theoretically take into account infinite levels and the transitions between them. Computationally, this is intractable.

The solution proposed in that 2006 publication (which marks a paradigm shift) was to divide the atom's levels. The lower levels (those from which the observed radiative emission typically originates) are kept exact and discretized, and all the infinite upper levels, up to the ionization limit, are grouped (bundled) into a single mathematical matrix that condenses their properties. The effects of these "bundled" upper levels are then algebraically projected onto the lower levels in what is called the **Bundle Model**. This technique provides an extremely robust description of the atom, capable of functioning in high and low density, high and low temperature regimes, and including all collisional processes in a computationally sustainable way.

1.24.1 The Contribution Function Equation

Without delving prematurely into the formal derivation, I present to you the key equation that unites theory with spectroscopic observations. This equation describes the intensity of the observed radiation:

$$I = \frac{1}{4\pi} \int G(T, N_e) \cdot \mathrm{DEM}(T), dT \tag{1.2}$$

Where:

- I represents the **Intensity** (the observable measured by the satellite).
- The $\frac{1}{4\pi}$ term derives from the solid angle for an isotropic emission.
- $G(T, N_e)$ is the **Contribution Function**. This is the theoretical heart of the system: it encompasses all the atomic modeling, the excitation/de-excitation rates, the transition probabilities, and the ionization functions. It strictly depends on temperature (T) and electron density (N_e).
- $\mathrm{DEM}(T)$ is the unknown term of the integral, and is what we wish to derive.

1.24.2 The Differential Emission Measure (DEM)

The **DEM (Differential Emission Measure)** function is the ultimate goal of our macroscopic modeling. The name itself indicates its nature: it is a measure of differential emission. Physically, it represents the spatial distribution of the plasma as a function of its temperature (and density). By inverting the integral equation shown earlier (noting the observed intensity I and calculating the contribution function G), the DEM is obtained. Knowing the DEM means having a complete mapping of the thermal structure of the plasma along the line of sight.

1.25 The Consequences of an Incorrect Model: The Case of Oxygen IV

I want to tell you an anecdote from my doctoral period that illustrates why a rigorous model (like the Bundle Model) is fundamental.

I was calculating the thermal profiles using data from the SOHO satellites (SUMER and CDS instruments). SOHO is located at the **Lagrange Point L1** (an unstable point of gravitational equilibrium between the Earth and the Sun, located at about 1.5×10^6 km from Earth. A satellite at L1 orbits the Sun keeping itself always aligned with the Earth, moving in a plane perpendicular to the Earth-Sun axis, allowing uninterrupted solar observation). Incidentally, the data was cross-referenced with that of the Japanese *Hinode* satellite, inserted in a terrestrial polar orbit.

The subject of my study were the **Oxygen IV (O IV)** lines, used specifically as plasma thermometers. By calculating the ratio between these lines using the complete model (the Summers model), I obtained consistent temperatures. At the same time, another research group published an article in which they reported incredibly high and physically inconsistent temperatures for the same solar region. Having obtained correct results myself, we investigated the origin of their error.

The problem lay precisely in the atomic model they adopted. They had calculated the ratios between two lines originating from the same upper energy level to two different lower levels (or originating from two different upper levels decaying to a common level). However, they had used the **basic Coronal Model**. This model ignored the redistribution of level populations induced by density and neglected the contributions of the non-bundled upper energy levels. Consequently, their model predicted a purely **linear** trend for the decay rates, heavily overestimating the theoretical intensities and, as a reflection, calculating dreadfully wrong temperatures (e.g., 10^6 K instead of 10^5 K).

This example clarifies the critical risk: the use of imprecise atomic physics will lead you to "discover" non-existent phenomena (false positives) and to draw completely wrong astrophysical conclusions.

1.26 Temporal Dynamics: Solar Flares

As a very last point, spectroscopy does not only produce static images, but allows us to follow the **temporal** evolution of space events.

Instead of acquiring global images, an instrument can acquire a spectrum at a single point with a very high temporal cadence. Analyzing the trend of **density as a function of time** during a **solar flare**, a peculiar behavior is noted. The density does not increase linearly and then decrease; the graph of intensity (and therefore density) as a function of time shows a series of **successive and overlapping peaks**. The curve rises "in steps" (up and up and up) and then gradually drops (down, down, down). The physical interpretation of this spectroscopic signature is that a flare is not a single, simple isolated explosion. On the contrary, the event is caused by a progressive ("chain") activation of interconnected magnetic loops, which explode in rapid sequence one after the other. This level of dynamic understanding is exclusively reachable through high-resolution spectroscopic analysis.

1.27 Advanced Modeling and Simulation Codes

This is the fundamental diagnostic tool at our disposal: spectroscopy allows us to decipher otherwise inaccessible phenomena. From an operational point of view, we extract the spectrum, run our calculation *routines*, obtain the ratio between the line intensities, and, starting from these data, we can apply different types of modeling.

Among the most complex models are the **Magnetohydrodynamics (MHD)** codes. These are advanced softwares that solve fluid dynamics equations coupled with electromagnetic field equations. Our task is to insert into these MHD codes the correct models, derived from atomic physics and plasma spectroscopy, so that the simulations produce physically valid results.

I am sure that some of you will use these tools in the future, especially those who decide to specialize in solar physics, where the use of such codes is extremely widespread. However, in astrophysics in general, the use of these simulations is a mandatory step. Unlike basic theory, in astrophysics we almost never have the luxury of observing a single isolated reaction coming from a precise point in space; we must therefore rely on hybrid modeling. In specific scenarios, we find ourselves analyzing a **non-equilibrium plasma**. To describe it correctly, it was necessary to couple two approaches:

- A density-dependent model, from the point of view of atomic physics.
- A fluid dynamic model, to describe the kinematic and dynamic evolution of the system.

1.28 Universality of Models: From Nuclear Fusion to Astrophysics

Exactly the same modeling framework is applied in the field of controlled thermonuclear fusion. Referring to what was previously shown about the Tokamak reactor, we can analyze the **divertor** region. The lower region of the divertor is relatively colder and presents thermodynamic properties similar to those of the solar atmosphere.

In this context, we sampled the density (N_e) and temperature (T) values for every single spatial point, calculating the local emission and simulating the propagation of radiation in the case of an equilibrium plasma. The crucial aspect is the following: once a solid and robust physical model is developed, it becomes universal. The same apparatus can be applied to countless environments and disciplines:

- Cosmology.
- Star formation.
- Solar physics.
- Interstellar medium.
- Plasmas induced by high-power lasers.
- Controlled thermonuclear fusion.

1.28.1 The Problem of Line of Sight Integration

Another peculiar challenge in astrophysics concerns the overlapping of emission. In space, the actual radiation we capture does not come from a two-dimensional layer, but can originate from portions of plasma with varying densities and temperatures along the entire optical path.

During modeling, it is necessary to sum all the emissive contributions along the **line of sight** to calculate what the resulting final radiation will be. In astrophysics we never see the single source point, but we exclusively perceive the integrated emission. An excellent diagnostic exercise consists precisely in the reverse process: to "disentangle" this information to trace back to the local properties of the plasma.

As a demonstration of this versatility, in my research work I compared the application of the same model on two radically different scenarios: solar flares and confined plasmas inside the Tokamak, obtaining excellent results in both cases.

1.28.2 The Educational Objective: Skill Development

The fundamental message I would like to convey to you today is that knowing the theoretical physical principles is not enough; one must know how to concretely translate these principles into model development. Physically building the model is an arduous task. The main technical skill you will acquire will be to intimately understand modeling: how to structure it and how to translate it into code. Once you have mastered this tool, you will be able to investigate what happens in the Sun, in the Universe, or in any other plasma environment.

1.29 Course Logistics, Tools, and Teaching Material

We conclude with some logistical notes for the upcoming lectures.

1.29.1 Teaching Method for the Upcoming Lectures

In the next module we will tackle the population balance equations and the radiative transfer equations. Due to the high density of mathematical formulas and analytical steps, the exclusive use of slides would be ineffective. Therefore, I will make use of a tablet to write and develop the derivations step by step in real time. I will share this digital notebook with you (for example via OneNote) so that you can keep track of everything we write. If I manage to complete the drafting of my handwritten notes from last year, I will also provide you with those at the end of the course.

1.29.2 Reference Texts (Textbooks)

Regarding textbooks:

- The reference texts are listed in the official syllabus.

- I will take care of explicitly linking the various books (or specific chapters) to the relevant sections of the syllabus, so you will know exactly where to study each topic.

- I strongly suggest you consult the texts. It is not necessary to buy them all; you can go to the library and photocopy or consult the sections of interest. Although I will provide comprehensive explanations, the support of an academic text is essential for deep learning.

Finally, an invitation to constructive collaboration: I am human and can make calculation errors or oversights during mathematical demonstrations. Please intervene openly and correct me without hesitation should you notice any inaccuracies.

We will stop here for today, this is the conclusion of the first lecture. See you at the next meeting.

Chapter 2

Lecture 2 - 05-10-2026

2.1 Reference Texts and Study Approach

This textbook is quite difficult, yet it is extremely complete and exhaustive. To be honest, I only discovered its existence during the third year of my degree course. Currently, I have deposited a copy of the volume at the library of the university school (the Scuola Superiore di Catania, to be precise). I warmly invite you to check its availability in the library, as it was the text on which I prepared for the first time on these topics.

During the lecture, I will adopt a blackboard approach, physically writing notes, as if I were in your place. This is because my intent is to proceed following the same approach I would adopt if I were a student myself.

2.2 Review of Quantum Mechanics

For today's lecture, I know well that you have all already taken exams regarding Quantum Mechanics (or, in some bachelor's degree systems, related courses within Mathematical Methods of Physics or Structure of Matter). However, I consider it essential to recall some fundamental concepts that will constitute the theoretical framework of plasma spectroscopy. We will start directly.

2.2.1 Physical Observables

The starting point is to define the quantities necessary to describe our physical system. From a physical point of view, we need mathematical entities capable of quantifying measurable properties.

What is measured experimentally is defined as an **observable**. An observable is a specific number that expresses the value of a **dynamic variable**. Classic examples of dynamic variables are:

- **Position** (spatial coordinates).
- The quantity of motion or **momentum**.

I do not think you have difficulties with either the nomenclature or the underlying physical concept.

2.2.2 The Linear and Hermitian Operator

In quantum mechanics, every observable is mathematically represented by an **operator**. The operator acts on the **state of the system**.

The state of the system is typically described by a mathematical function to which we assign a Greek letter, commonly ψ (psi) or ϕ (phi). Choose the notation you prefer, the important thing is consistency.

An operator is, in fact, a mathematical rule that, applied to a specific state function ψ, converts it into a new function ψ'. The fundamental eigenvalue equation is written in the form:

$$\alpha\psi = a\psi \tag{2.1}$$

Where:

- α represents the operator (which we will call **linear Hermitian operator**).
- ψ is the eigenfunction.
- a is the number (scalar) representing the eigenvalue corresponding to the measured observable.

There are various notational conventions for operators. Instead of using Greek letters like α, you might find operators indicated with capital Latin letters endowed with a circumflex accent (the *hat* or *cap*), for example:

$$\hat{A}\psi = a\psi \tag{2.2}$$

Since there is no unequivocal notation in the literature, we will use the forms that will progressively prove more convenient, provided the context is clear.

2.2.3 Properties of Operators: Linearity

Not having at hand the exact syllabus of your bachelor's degree course, let's quickly review what it means that an operator is **linear**.

An operator A is defined as linear if, given two generic functions ψ_r and ψ_s and a constant a, the following distributive property holds:

$$A(a\psi_r + a\psi_s) = aA\psi_r + aA\psi_s \tag{2.3}$$

In the most general case, with different constants a and b associated with the respective functions, the linearity condition for an operator α is strictly expressed as:

$$\alpha(a\psi_r + b\psi_s) = a\alpha\psi_r + b\alpha\psi_s \tag{2.4}$$

2.2.4 Properties of Operators: Hermitianity

The second fundamental property is **Hermitianity**. An operator A is defined as **Hermitian** if it coincides with its adjoint operator. The adjoint operator is usually indicated with a dagger-shaped superscript (†, commonly read as "dagger" in English). Therefore, the condition of Hermitianity is written:

$$A = A^\dagger \tag{2.5}$$

These are basic concepts, which you have almost certainly encountered and applied numerous times, but which are essential to have firmly established in your memory to proceed.

2.3 Eigenvalues, Eigenstates and Physical Measurements

We introduce an eigenvalue α', corresponding to which we can write the following equation:

$$\alpha\psi = \alpha'\psi \tag{2.6}$$

This is exactly the same equation introduced previously, written in a different way. The goal here is to get you used to using different letters and notations to indicate the same concept, so that you can be more flexible and recognize the physical structure regardless of the formalism adopted.

The parameter α' (the eigenvalue) can be:

- A **discrete number** (e.g., $1, 2, 3$).
- A **continuous** value (therefore represented as a function).

All the functions ψ that satisfy this particular equation for a given operator are called **eigenstates**. For the eigenstates ψ that satisfy the equation with respect to a specific α', we formally say that they "belong to the eigenvalue α'". Often, to make this link explicit, we will indicate them with the indexed notation:

$$\alpha\psi_{\alpha'} = \alpha'\psi_{\alpha'} \tag{2.7}$$

There is another point of crucial importance. Since the outcome of a real physical measurement must necessarily provide a result belonging to the field of real numbers for every possible state ψ, it follows that **all physically measurable quantities are represented by real eigenvalues**. I emphasize this aspect because the quantum formalism appears deeply theoretical and abstract. When performing calculations on these mathematical objects, it is easy to lose their practical meaning; however, always remember that these *real observables* translate into direct experimental measurements: momentum, position, and energy.

2.4 The Operator Spectrum and Continuity

Here is the fundamental element: the complete set (the full set) of all eigenvalues of an operator α is called the **Spectrum** of α.

This concept provides you with the key to connecting abstract mathematics to what we discussed last time regarding the intensity of the electromagnetic spectrum. We said that the spectrum essentially manifests itself in two ways:

- Discrete: originating **spectral lines**.
- Continuous: originating a continuous spectrum.

Let us consider an observable (which from now on we will indicate with the letter ω) whose set of eigenvalues is not purely discrete, but turns out to be **entirely or partially continuous**. In this case, the eigenstates ψ will belong to values (let's call them ω') positioned in a continuous interval and not on discrete levels.

In a continuous spectrum, the normalization condition between the states $\psi_{\omega'}$ and $\psi_{\omega''}$ can no longer make use of the Kronecker Delta, but must be rewritten in this way:

$$\int \psi^*_{\omega'} \psi_{\omega''}, d\tau = \delta(\omega' - \omega'') \tag{2.8}$$

The function on the right-hand side is the **Dirac Delta Function** ($\delta(x)$). This is a particular mathematical distribution, defined by the following properties (setting for simplicity $x = \omega' - \omega''$):

$$\delta(x) = \begin{cases} 0 & \text{per } x \neq 0 \ +\infty \\ \text{per } x = 0 \end{cases} \tag{2.9}$$

Subject to the integral normalization condition:

$$\int_{-\infty}^{+\infty} \delta(x), dx = 1 \tag{2.10}$$

2.5 Representations: The Approach of Schrödinger and Dirac

In quantum mechanics, there are various formal representations to describe exactly the same phenomenon. One can use the differential formulation, matrices, and so on. They all converge to the same result: describing the evolution of eigenstates and eigenvalues under the action of operators.

Let's proceed to review these representations to gradually reintroduce the variables that will be necessary for us.

2.5.1 From the Wave Function to Bra-Ket Notation

Let's focus on the classical Schrödinger Representation applied to vector notation. Consider a standard integral in wave mechanics, the calculation of the expectation value (or matrix element):

$$\int \psi^* A \psi, d\tau \tag{2.11}$$

How can we condense this information? By resorting to the **Bra** and the **Ket** (the notation introduced by Paul Dirac).

- The column vector, which represents the standard wave function ψ, is called **Ket** and is indicated as $|\psi\rangle$.
- The complex conjugate row vector, ψ^*, is called **Bra** and is indicated as $\langle\psi|$.

When they are joined around an operator A, we obtain the formulation:

$$\langle\psi| A |\psi\rangle \tag{2.12}$$

This notation, which may appear hermetic at first glance, is **exactly equivalent** to the integral $\int \psi^* A\psi \, d\tau$. The only difference lies in the fact that the spatial dependence of the function $\psi(x)$ or its complex conjugate $\psi^*(x)$ inside the integral is not made explicit; the functions are treated directly as vector or matrix entities (state vectors).

If you try to insert the dependence on a variable, for example on the coordinates x and time t, this equivalence will be even clearer to you. The mathematical step is the same one you will have already encountered when tackling basic QM courses. This serves solely as a reminder, to standardize the nomenclature and guarantee that everyone understands the formalism I will adopt in class, sparing you from having to hurriedly pore over old notes in case of temporary amnesia.

2.6 The Rules of Quantum Mechanics

Having defined the notation, let's get to the heart of the matter by analyzing the fundamental principles of quantum mechanics applied to kinematic and thermodynamic variables.

2.6.1 Coordinates and Conjugated Momentum

Let us consider a Cartesian coordinate system. To generalize and extend the concept to N dimensions, we will indicate the coordinates as $x_1, x_2, \ldots, x_{3N}$ (rather than using x, y, z). Concurrently, we define the associated **conjugated momentum**, indicated with $p_1, p_2, \ldots, p_{3N}$.

These two quantities (coordinates and conjugated momenta) constitute the backbone of kinematics, and in the quantum regime they obey precise **Commutation Rules**. Let us recall the definition of the **commutator** between two operators A and B:

$$[A, B] = AB - BA \tag{2.13}$$

For position and momentum operators, the rules are explicated as follows:

- Commutator between the spatial components:

$$[x_i, x_j] = x_i x_j - x_j x_i = 0 \tag{2.14}$$

- Commutator between the conjugated momenta:

$$[p_i, p_j] = p_i p_j - p_j p_i = 0 \tag{2.15}$$

- Crossed commutator between position and momentum (formal Heisenberg Uncertainty Relation):

$$[x_i, p_j] = x_i p_j - p_j x_i = i\hbar\delta_{ij} \tag{2.16}$$

(Where i is the imaginary unit and $\hbar$ is the reduced Planck constant).

2.6.2 The Hamiltonian and Time Evolution

Exactly in analogy with the total energy of a system in classical mechanics, in quantum mechanics we define the **Hamiltonian function**, commonly indicated with the letter H.

The Hamiltonian is a function depending on both the momenta p and the coordinates x. While in classical physics it provides Hamilton's equations of motion, in quantum theory the Hamiltonian operator deterministically determines the evolution of the system's state over time.

Resuming the formal structure (replacing the generic operator α used so far with the Hamiltonian H), the equation of motion is derived by defining the derivative of the quantum state with respect to time. This is the very famous **time-dependent Schrödinger Equation**:

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi \tag{2.17}$$

When we apply the Hamiltonian operator H to a specific stationary state (an eigenstate of the system), it returns the energy eigenvalue E of that state, i.e., the energy associated with the photon emitted or absorbed during the transition. The stationary (time-independent) form is reduced to:

$$H\psi = E\psi \tag{2.18}$$

The importance of the Hamiltonian lies in the fact that it is the observable responsible for quantifying energy. As the state evolves over time and the wave function changes phase, it is the application of the H operator that governs and quantizes the energy levels associated with the studied plasma and radiation.

2.7 Time Evolution and Stationary States

Resuming the time dependence discussed previously, the time-dependent Schrödinger equation governs the evolution of the state of the system:

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t) \tag{2.19}$$

(Where H is the Hamiltonian operator, in charge of defining the energy, and the constant $\hbar = h/2\pi$).

The time evolution of a state (for systems with a Hamiltonian not explicitly dependent on time) produces a phase factor. The time derivative leads to the exponential:

$$e^{-i\frac{E}{\hbar}t} \tag{2.20}$$

The complex conjugate state ψ^* will exhibit the opposite phase factor:

$$e^{+i\frac{E}{\hbar}t} \tag{2.21}$$

To clarify how the system evolves and in what way the Hamiltonian defines the energy of the quantum state, let us consider the calculation of the expectation value of a generic observable. Integrating over space:

$$\int \psi^*(x,t)A\psi(x,t), d\tau \tag{2.22}$$

Substituting the explicit time dependence into the states:

$$\int \left(\psi^*(x,0)e^{+i\frac{E}{\hbar}t}\right) A \left(\psi(x,0)e^{-i\frac{E}{\hbar}t}\right) d\tau \tag{2.23}$$

Since the operator A (if it does not depend on time) acts only on the spatial part $\psi(x,0)$, the complex exponentials with opposite phase cancel each other out ($e^{+i\frac{E}{\hbar}t} \cdot e^{-i\frac{E}{\hbar}t} = 1$).

The result shows us that the value of the observable measured at time t (final state) is **identical** to the value measured at time $t = 0$ (initial state). When this happens, we are in the presence of what is defined as a **Stationary State**, that is, a state in which the probabilities of the physical quantities do not change over time. The equation that describes these states is the time-independent Schrödinger Equation:

$$H\psi = E\psi \tag{2.24}$$

This is the "friendly" formula that we will encounter and use countless times to derive the energy levels and explain the emission of photons.

2.8 The Hydrogen-like Spectrum: The Single-Electron Model

As mentioned during the introduction to plasma spectroscopy in the previous lecture, let us start from the most basic case (which many of you will have already encountered in Structure of Matter): the **single-electron** atomic spectrum. This does not exclusively include the hydrogen atom (H), but embraces an entire family of ions called **Hydrogen-like Ions** (Hydrogen-like atoms or H-like ions).

Hydrogen-like ions are atoms of heavier elements (present in the periodic table) that have been progressively ionized, losing their electrons until they are left with **only one electron** orbiting around the nucleus. Since their structure is reduced to a nucleus and a single electron, their physical behavior and the equations that govern them are extremely similar to those of hydrogen, although the Coulomb attraction is greater due to the higher number of protons in the nucleus.

A classic example: **Helium** (He). The neutral atom has 2 electrons. If it loses one electron, it becomes the singly ionized Helium ion. Since it has only one electron remaining, it behaves like a hydrogen-like ion.

2.8.1 Spectroscopic Notation and Ionization

There is a marked difference in notation between chemistry and astrophysical spectroscopy for indicating ions.

- **Chemical Notation**: The ionization state is indicated by placing "plus" signs as superscripts after the element symbol. Singly ionized Helium will be written as He^+.
- **Spectroscopic Notation**: In spectroscopy, the neutral atom is not indicated with a zero, but the count starts from "I" (one in Roman numerals). The ionized state scales accordingly:

- Neutral atom = I (e.g., H I, He I)
- Singly ionized = II (e.g., He II, corresponding to He^{+})
- Doubly ionized = III (e.g., Li III, corresponding to Li^{2+})

For example, **Lithium** (Li), which basically possesses 3 electrons (Z = 3), to become hydrogen-like must lose 2 electrons, becoming Li^{2+} in chemical notation, which will correspond to **Li III** in spectroscopic notation.

This Roman numeral system for spectroscopy can initially be confusing, as the chemical exponent of the charge and the spectroscopic index are out of phase by 1 unit ($n_{\text{spectroscopic}} = n_{\text{charge}} + 1$). I urge you to always keep a **Periodic Table** at hand during these lectures. Knowing the atomic numbers Z of every element by heart is not a course requirement; the periodic table is the basic tool of our trade.

2.9 The Hydrogen-like Atom Model: Nucleus-Electron Interaction

Let us proceed to physically model our hydrogen-like system. The components of our basic model are exclusively two:

1. A massive **Nucleus**, with a net positive charge equal to $+Ze$ (where Z is the atomic number, which for neutral hydrogen is $Z = 1$, and e is the elementary charge).
2. A **Single Electron**, with mass m_e and charge $-e$.

The two particles interact solely via an electrostatic attractive force: the **Coulomb Force**. Neglecting the vectorial formalism for a moment, the magnitude of the Coulomb force is expressed by Coulomb's law:

$$F_C = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(e)}{r^2} \tag{2.25}$$

Where r is the radial distance between the nucleus and the electron. As is evident from the equation, the force is proportional to the product of the charges and inversely proportional to the square of the distance.

2.9.1 The Reduced Mass and the Two-Body Problem

The nucleus is not fixed in space; both the nucleus (with mass m_p, meaning for simplicity the nuclear mass) and the electron (with mass m_e) move around their common center of mass. This is the classic two-body problem.

To simplify the analysis and reduce the problem to the motion of a single virtual "particle" around a fixed center, in mechanics the concept of **Reduced Mass** is introduced, usually indicated with the letter μ:

$$\mu = \frac{m_e \cdot m_p}{m_e + m_p} \tag{2.26}$$

To understand how much the reduced mass is worth, let us recall the value of the electron mass. The electron has a mass $m_e \approx 9.1 \times 10^{-31}$ kg, which

expressed in rest energy is equivalent to ~ 0.511 MeV. The mass of the proton (or of the nucleus) is immensely greater (about 1836 times greater for hydrogen). Since $m_p \gg m_e$, we can divide the numerator and denominator of the reduced mass equation by m_p:

$$\mu = \frac{m_e}{1 + \frac{m_e}{m_p}} \tag{2.27}$$

Since the term m_e/m_p tends to zero, the equation is simplified and the reduced mass coincides in first approximation with the mass of the electron:

$$\mu \approx m_e \tag{2.28}$$

This approximation widely justifies the basic theoretical assumption in which the nucleus is considered stationary (at the center of the coordinate system) and the orbit of only the electron around it is modeled.

2.9.2 The Central Force Field and the Electrostatic Potential

Following the mass approximation, the electron can be modeled as a particle in motion inside a **central force field** (Central Field, corresponding to the "centrally symmetric field" of analytical mechanics).

In a central field, the potential energy U (or V) does not depend on the angular direction, but is an exclusive function of the radial distance r from the force center (the nucleus). Therefore we will write:

$$U = U(r) \tag{2.29}$$

The electrostatic potential energy for the hydrogen-like system, deriving from the integration of the attractive Coulomb force, has the well-known form:

$$U(r) = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} \tag{2.30}$$

The negative sign indicates that the potential is attractive: the electron is "bound" to the nucleus (forming a potential well) and energy must be supplied to the system to separate the two charges to infinity.

2.10 The Single-Electron Atom Model

To define our basic model for the single-electron atom (or hydrogen-like ion), we must necessarily introduce some initial approximations. So as not to excessively complicate the discussion, we will assume the following:

- **We neglect the electron spin** (and the relative electromagnetic interactions associated with it).
- **We neglect relativistic effects**.

Under these hypotheses, we can define a relatively simple Hamiltonian (H) to describe our atom. Starting from the classical formulation and passing to quantum mechanics, the Hamiltonian is expressed by the sum of kinetic energy

and potential energy (U). In terms of the momentum components (p_x, p_y, p_z), we will have:

$$H = \frac{p_x^2 + p_y^2 + p_z^2}{2\mu} + U \tag{2.31}$$

That is, in a more compact form:

$$H = \frac{p^2}{2\mu} + U \tag{2.32}$$

This quantum expression describes the electron exploiting the reduced mass μ that we have previously defined.

2.10.1 Angular Momentum and Commutation Rules

In addition to momentum, we must consider the **angular momentum L**. It is a quantity that you have already encountered in the basic Physics 1 courses. Angular momentum has three directional components: L_x, L_y, L_z, to which the squared magnitude operator L^2 is added. (Notational note: spatial coordinates can be indifferently indicated as x, y, z or as x_1, x_2, x_3, the choice is irrelevant for the physics of the problem).

In quantum mechanics, the fundamental aspect lies in the commutation properties between the Hamiltonian H, the squared magnitude of the angular momentum L^2, and the individual components of **L**. It is not possible to simultaneously know all the directional components of angular momentum, but it is possible to determine H, L^2 and *only one* of the spatial directions (conventionally L_z is chosen).

This implies that the commutators between these quantities are zero:

$$[H, L^2] = 0 \tag{2.33}$$

$$[H, L_z] = 0 \tag{2.34}$$

$$[L^2, L_z] = 0 \tag{2.35}$$

Since these operators commute with each other, what we are looking for are the **simultaneous eigenstates** of H, L^2 and L_z.

2.11 Schrödinger Equation and Laplacian Operator

To find these eigenstates, we move to the **Schrödinger Representation**. The stationary Schrödinger equation is written as:

$$-\frac{\hbar^2}{2\mu}\nabla^2\phi + U\phi = E\phi \tag{2.36}$$

Allow me a digression on notation, as I noticed that last year it generated confusion. The symbol ∇^2 (nabla squared) appears in the equation. In some texts, or from other professors, you might see it indicated with a triangle, namely Δ. Both ∇^2 and Δ represent the same operator: the **Laplacian operator**. They are mathematically identical formulations, it is just a matter of calling or writing them differently. In this course we will use the notation with nabla squared (∇^2). In the above equation, ϕ represents the state wave function, U is the potential energy and E is the energy eigenvalue.

2.11.1 Insertion of the Coulomb Potential

How do we define the potential energy U in the Coulomb field? As you know from Physics 1, potential energy derives from force. Since the Coulomb force is inversely proportional to the square of the distance ($1/r^2$), the associated (attractive) potential energy is proportional to $1/r$.

Let us therefore substitute the explicit expression of the potential $U(r)$ inside our Schrödinger equation:

$$-\frac{\hbar^2}{2\mu}\nabla^2\phi + U(r)\phi = E\phi \tag{2.37}$$

At this point, we can proceed with a reorganization of the terms. We bring the term $E\phi$ to the left member to equate the equation to zero and, if necessary, change the signs by factoring out the wave function ϕ. We will have:

$$\left(-\frac{\hbar^2}{2\mu}\nabla^2 + U(r) - E\right)\phi = 0 \tag{2.38}$$

In this equation, r represents the radial distance between the electron and the nucleus. If we want to specialize the equation for the neutral hydrogen atom right away, we will set the atomic number $Z = 1$ inside the expression for $U(r)$.

At this point, with the differential equation correctly set up, we will take a short 5-minute break before proceeding with its resolution. If there are questions up to here, I am available.

2.12 Resolution of the Schrödinger Equation in Polar Coordinates

Our goal, now, is to solve the differential Schrödinger equation just set up to find the eigenvalues of energy, angular momentum and the associated wave functions.

As you probably know from mathematical analysis and quantum mechanics, to easily solve this type of equations with centrally symmetric potentials (where U depends only on r), it is necessary to abandon Cartesian coordinates and move to **spherical polar coordinates** (r, θ, φ). This change of coordinates is crucial because it allows the **separation of variables**.

We will not carry out all the intermediate analytical steps today, as the focus of this lecture will be on the construction of electron configurations (a more practical but fundamentally important exercise). Let's rewrite the Laplacian operator ∇^2 in spherical polar coordinates to show the structure of the equation:

$$\left[-\frac{\hbar^2}{2\mu}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right) - \frac{Ze^2}{r}\right]\phi = E\phi \tag{2.39}$$

(Note: the professor indicates $U = -\frac{Ze^2}{r}$ in Gaussian CGS units, omitting the $1/4\pi\varepsilon_0$ factor for compactness, as is typical in the astrophysical/spectroscopic field).

2.12.1 Separation of Variables and Factorization of the Wave Function

Exploiting the symmetry of the system, the solution $\phi(r, \theta, \varphi)$ can be factorized into the product of two independent functions:

$$\phi(r, \theta, \varphi) = R_{nl}(r) \cdot Y_{lm}(\theta, \varphi) \tag{2.40}$$

Where:

- $R_{nl}(r)$ is the **radial part** of the wave function, which depends solely on the distance from the nucleus r. Its functional form is linked to the Laguerre polynomials.
- $Y_{lm}(\theta, \varphi)$ is the **angular part**, which depends on the angular coordinates θ (colatitude) and φ (azimuth). These functions are mathematically known as **Spherical Harmonics**.

(A personal note: historically I have always drawn the Y symbol of spherical harmonics starting from the opposite direction compared to the norm. No one ever corrected me until university, so forgive me if my handwriting of the "Y" on the blackboard appears unusual!).

2.13 Stationary States and Eigenvalues

This mathematical decomposition returns exactly the **stationary states** we were discussing earlier. These states are simultaneous eigenfunctions of the operators H, L^2, and L_z. Applying these operators to the angular part Y_{lm} (or to the complete wave function ϕ), we obtain the respective eigenvalues:

1. **Squared magnitude of the angular momentum (L^2)**:

$$L^2\phi = l(l+1)\hbar^2\phi \tag{2.41}$$

   (One is often tempted to say that the eigenvalue of L^2 is simply $l^2\hbar^2$, but the rigorous quantum derivation imposes the factor $l(l+1)$).

2. **Z component of the angular momentum (L_z)**:

$$L_z\phi = m\hbar\phi \tag{2.42}$$

2.13.1 Quantum Numbers

The equations just written introduce three fundamental numbers that govern atomic structure. They are the **quantum numbers**, which you have certainly studied in chemistry:

- n: **Principal quantum number**. Determines the energy level and the size of the orbital.

$$n = 1, 2, 3, \ldots, \infty \tag{2.43}$$

“‘

- l: **Angular momentum quantum number** (or azimuthal/orbital). Defines the shape of the orbital. It is strictly limited by the value of n:

$$l = 0, 1, 2, \ldots, (n-1) \tag{2.44}$$

- m (or m_l): **Magnetic quantum number**. Defines the projection of the angular momentum (the orientation of the orbital) along the z-axis. It is limited by the value of l:

$$m = 0, \pm 1, \pm 2, \ldots, \pm l \tag{2.45}$$

(It therefore assumes $2l + 1$ integer values between $-l$ and $+l$).
‘‘

2.14 Energy Eigenvalues and Spectroscopic Units of Measurement

The resolution of the radial part of the Schrödinger equation provides us with the eigenvalues for the Hamiltonian operator, that is, the **energy levels** of the hydrogen-like system:

$$E_n = -\frac{\mu Z^2 e^4}{2\hbar^2 n^2} \tag{2.46}$$

These eigenvalues correspond exactly to those derived in Bohr's semi-classical theory.

In spectroscopy, it is standard practice to group the constant terms into dedicated units of measurement to streamline calculations.

- **Atomic unit of energy**: the block of constants $\frac{\mu e^4}{\hbar^2}$ represents the atomic unit of energy (linked to the Hartree).
- **Rydberg energy (Ry)**: is defined as the binding energy of the ground state of hydrogen.

$$Ry = \frac{\mu e^4}{2\hbar^2} \approx 13.6 \text{ eV} \tag{2.47}$$

Therefore, the energy levels can be rewritten in the compact form:

$$E_n = -Ry\frac{Z^2}{n^2} \tag{2.48}$$

2.14.1 Wavenumber

In the spectroscopic field, the energy of transitions is very often measured not in Joules or eV, but in **wavenumbers**, conventionally indicated with the letter σ (or $\tilde{\nu}$). The wavenumber is defined as the reciprocal of the wavelength (λ):

$$\sigma = \frac{1}{\lambda} \tag{2.49}$$

It is generally expressed in cm^{-1}. This value is directly proportional to the energy of the photon transition ($E = hc\sigma$) and is the standard operational parameter in spectroscopic tables.

2.14.2 Ionization Potential

Before moving on to the practical part, let us formally define the **Ionization Potential** (E_i or IP). Ionization energy is the energy required to tear the electron from its ground state and bring it to an infinite distance from the nucleus (where the potential energy is zero). Mathematically it is the difference between the energy at infinite distance ($n = \infty$) and the energy of the ground state ($n = 1$):

$$E_i = E_\infty - E_1 = 0 - \left(-Ry\frac{Z^2}{1^2}\right) = Ry \cdot Z^2 \tag{2.50}$$

For the hydrogen atom ($Z = 1$), the ionization potential coincides exactly with the Rydberg constant (13.6 eV).

2.15 Electron Configuration and Orbital Nomenclature

At this point, having introduced the quantum numbers, we possess the tools to trace the structure of energy levels, not only of the hydrogen-like atom, but of all multi-electron atoms.

The **Electron Configuration** (Configuration of an electron) is uniquely defined by the pair of quantum numbers:

$$(n, l) \tag{2.51}$$

There is no need to specify m to define the isolated energy level or sublevel (in the absence of external magnetic fields that remove degeneracy).

Furthermore, it is a historical convention (deriving from the first spectroscopic observations) to associate the angular quantum number l not with a number, but with a specific **letter of the alphabet**, thus defining the **orbital**. The sequence of letters as a function of the value of l is as follows:

- $l = 0 \longrightarrow$ **s**
- $l = 1 \longrightarrow$ **p**
- $l = 2 \longrightarrow$ **d**
- $l = 3 \longrightarrow$ **f**
- $l = 4 \longrightarrow$ **g**
- $l = 5 \longrightarrow$ **h**

The sequence continues in alphabetical order ($i, k, \ldots$ omitting the j). In astrophysical or chemical practice, you will hardly encounter ions that occupy orbitals with very high l in the ground state (however, in highly excited states typical of plasmas or in recombination lines, you will be able to observe transitions involving h orbitals or higher).

2.16 Orbital Filling Rules and Spectroscopic Nomenclature

It might seem like an academic detail, but it is essential to know that there are orbitals (like the h orbitals) which, although not populated in the ground state, provide a fundamental contribution in the complex atomic models that electrons can access in the excited states of a real plasma.

Let's operationally define an **orbital**. We have said that the configuration of an electron is determined by the principal quantum number n and the angular one l. Let's resume the simplest example:

- $n = 1$
- $l = 0$

The configuration of this orbital will be written: **1s**.

2.16.1 Practical Examples of Electron Configurations (Neutral Atoms)

Let's now do some simple practical exercises to refresh your memory on how configurations are built. Arm yourselves with your periodic table. We will start with neutral atoms.

Let's recall two fundamental rules:

1. Every orbital (or rather, every sublevel defined by l) can host a maximum number of electrons, dictated by the Pauli Exclusion Principle ($2(2l + 1)$ electrons).
    - **s** sublevel ($l = 0$): maximum **2** electrons.
    - **p** sublevel ($l = 1$): maximum **6** electrons.
    - **d** sublevel ($l = 2$): maximum **10** electrons.
2. The superscript next to the orbital letter indicates the number of electrons contained in it.

Let's proceed with the list of elements:

1. **Hydrogen (H)**
    - Atomic number $Z = 1$ (1 proton, 1 electron).
    - Configuration: $1s^1$

    ‘‘‘

2. **Helium (He)**
    - Atomic number $Z = 2$ (2 protons, 2 electrons).
    - Configuration: $1s^2$ (The $n = 1$ level is now full).
3. **Lithium (Li)**
    - Atomic number $Z = 3$ (3 protons, 3 electrons).

- Configuration: $1s^2\,2s^1$

4. **Carbon (C)** and **Oxygen (O)** (Crucial elements in astrophysics)
   - Carbon ($Z = 6$): $1s^2\,2s^2\,2p^2$
   - Oxygen ($Z = 8$): $1s^2\,2s^2\,2p^4$

2.16.2 Digression: Atomic Number and Atomic Mass (Isotopes for Fusion)

Let's make a quick terminological parenthesis. The number Z represents the **Atomic Number**, that is, the number of protons in the nucleus. It should not be confused with the **Atomic Mass**, which is given by the sum of the number of protons and the number of neutrons ($A = Z + N$).

Standard hydrogen (whose nucleus is a single proton) has zero neutrons. But its **isotopes** exist:

- **Deuterium**: has 1 proton and 1 neutron.
- **Tritium**: has 1 proton and 2 neutrons (and is radioactive).

I mentioned deuterium and tritium because, in laboratory plasmas for controlled thermonuclear fusion, ordinary hydrogen is not used, but rather a mixture of deuterium and tritium, which is much more efficient for triggering fusion reactions.

2.16.3 Exceptions in the Filling Order: Iron (Fe)

Returning to configurations, let's look at **Iron (Fe)**, which has atomic number $Z = 26$. Following the Aufbau Principle (progressive construction), we fill the orbitals according to the scale of increasing energies:

- $1s^2 \quad 2s^2 \quad 2p^6 \quad 3s^2 \quad 3p^6$ (we have reached 18 electrons, Argon).
- At this point, there are 8 electrons left to place.

From a strictly energetic point of view, the $4s$ orbital fills *before* the $3d$ orbital because it has a slightly lower energy in the neutral ground state. We would therefore have: $4s^2\,3d^6$.

However, especially in **astrophysics and spectroscopy**, it is common practice to write the configuration keeping the levels ordered based on the principal quantum number n, regardless of energy. Therefore we will write:

$$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6, 4s^2 \tag{2.52}$$

Notice how the $3d$ orbital, which can hold up to 10 electrons, contains only 6, despite being followed by the full $4s$. Both notations (energetic order or order by principal quantum number) are acceptable for the purposes of this course, provided the count and the electronic assignment are exact. What you must assimilate is that the energetic order of the levels ($3d$ vs $4s$) **is not an absolute dogma**, but depends critically on the density, temperature, and ionization state of the plasma (energy levels crossing).

2.17 Electron Configurations of Astrophysical Ions

As reiterated previously, in space astrophysics (with the exception of cold nebulae) neutral matter is almost non-existent. One exclusively observes plasma, that is, ionized matter. We must learn to write the configurations of ions.

Let's consider an extreme case, much observed in solar flares: **Iron Twenty (Fe XX)**. This ion presents intense emission lines at extremely high temperatures.

- The symbol **Fe XX** (spectroscopic notation) indicates Iron ionized 19 times (charge +19).
- Neutral Iron has 26 electrons. Removing 19, **7** remain.

We are faced with an Iron nucleus ($Z = 26$) around which only 7 electrons orbit. Which neutral element has 7 electrons? **Nitrogen (N)**. We can therefore state that Fe XX belongs to the **isoelectronic sequence of Nitrogen** (Nitrogen-like ion). Fe XX behaves spectroscopically in a similar way to neutral Nitrogen, while not being identical at all. The difference lies in the nucleus: the Fe XX nucleus has 26 protons, exerting an enormously more intense Coulomb potential on the 7 electrons compared to the 7 protons of the Nitrogen nucleus. The energy of the levels will therefore be very different, but the structure of the valence electron configuration will be equivalent.

Let's write the configuration for Fe XX, placing its 7 electrons:

$$1s^2, 2s^2, 2p^3 \tag{2.53}$$

This simple calculation is the basis for defining the spectral lines that this ion will produce. All possible spectral terms and multiplets that make up the observable spectrum derive from this basic configuration.

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2.18 Essential Resources: The NIST Database

For anyone intending to analyze real spectroscopic data, it is indispensable to know and know how to query atomic databases. The global reference database for spectroscopy is the **NIST Atomic Spectra Database** (National Institute of Standards and Technology). I invite you to memorize its use or save the link. It collects the data of all spectral lines observed and measured in the laboratory, as well as theoretical models for atomic energy levels.

(Side note for the current context: The management of large international atomic databases is complex and dependent on state public funding. Historically these servers, including NIST and associated databases like NRL, relied on funds from American universities (e.g., Harvard) or government institutions (e.g., NASA). The recent drastic cuts to research budgets have caused delays in updates - for example, the latest updates visible on the portal date back to 2024 - or physical transfers of servers between various world research institutes. Fortunately, at the moment the service remains operational and is the pillar of our discipline, working in synergy with other immense open-source libraries developed mainly between the USA and the UK, such as the astrophysical database ***CHIANTI****).*

2.18.1 Example of a NIST Database Query

I will briefly show you how to extract data from the portal:

1. From the main page, select the "Lines" section.
2. The ionic species is requested. Enter the ion of interest **using spectroscopic notation**. Type, for example: `Fe XX`.
3. Select the **wavelength** range. In the astrophysical EUV or optical field, the convention is to use **Ångströms (Å)** (alternatively, nanometers or micrometers for the infrared).
4. Set the lower and upper limits of the investigation range.

Upon submitting the query (Retrieve Data), the system will return a table containing all the known transitions for that interval, specifying the relative intensity, the transition probabilities, and the exact configurations of the upper and lower levels between which the quantum leap occurs. This is the diagnostic tool with which you will inevitably have to familiarize yourselves.

2.19 Navigating the NIST Database

Let's pick up the discussion on the **NIST Atomic Spectra Database**. Once the query is executed for a given ion (for example Fe XX) in the desired wavelength range, the system returns a table structured in columns. In this phase, we will momentarily ignore some columns to focus on those fundamental for our analysis.

Here are the key columns you must know how to read:

- **Wavelength (λ)**: Usually expressed in Ångströms or nanometers, depending on the chosen settings.
- **Energy Levels**: This column shows the energies of the lower level and the upper level involved in the transition.
- **Configuration**: This is the crucial column for the topic covered today. It shows you the exact electron configuration of the state. For example, you might read $1s^2\,2s^2\,2p^3$ for the ground state of Fe XX, or excited configurations in which an electron has "jumped" to a higher level. (Note: often in databases for heavy ions, the inner closed "cores" are omitted for brevity, focusing on valence electrons).
- **Term** and **Total Angular Momentum (J)**: Columns that exactly define the quantum state, concepts we will address in detail later (e.g., triplet, singlet, ground or excited state).

In addition to spectral lines (Lines), NIST allows you to directly query **Energy Levels (Levels)**. Returning to the main menu, you can select "Levels" and enter your ion. In this form, you can customize the output:

- **Energy Unit**: You can choose between Wavenumbers (cm^{-1}), Electron volts (eV), Rydbergs, or frequency. For high-energy plasmas, the eV is often a compact and convenient choice.

- **Table Format**: It is possible to order the levels by increasing energy, grouping them based on the configuration and the spectral term.

It is fundamental to familiarize yourselves with this tool, since radiative transitions (emission/absorption) obey precise quantum **Selection Rules** that determine which "jumps" between an upper and a lower level are allowed or forbidden. This database condenses the entire theoretical scaffolding into directly usable data. If you search Google for "*NIST atomic database*", the first result will be the correct one.

2.20 Practical Exercise: Writing Configurations

Before closing, I leave you with three simple exercises on electron configurations, similar to the examples done a little while ago. The goal is to consolidate the concepts discussed today, in preparation for the next lecture. You can do them at home (they will require no more than 15 minutes) or we can check them together next time.

Write the ground electron configuration and identify the isoelectronic sequence for the following ions:

1. **Mg X** (Magnesium ionized 9 times). Data: $Z_{\mathrm{Mg}} = 12$.
2. **Si XII** (Silicon ionized 11 times). Data: $Z_{\mathrm{Si}} = 14$.
3. **S XIV** (Sulfur ionized 13 times). Data: $Z_{\mathrm{S}} = 16$.

(Note: pay close attention to the relationship between spectroscopic notation and the number of lost electrons, to calculate how many electrons actually remain on the orbit).

2.20.1 The Usefulness of Exercises vs Artificial Intelligence

You are obviously aware that, if you feed this request to an Artificial Intelligence, it will return the exact solution to you in a fraction of a second. But the purpose of this exercise is not to passively obtain the result; it is to *train reasoning*. Making the effort to derive the configuration manually, relaxing and applying the Aufbau rules, will allow you to internalize the atomic structure. AI will not take the exam for you nor will it interpret anomalous spectra for you in the laboratory. Take these 15 minutes of tranquility to solve them independently.

2.21 Logistical Communications and Next Lectures

Our next lecture will be held **next Monday**. As I informed you at the beginning of the course, this Thursday I will be out of town, therefore the lecture is canceled.

In the meantime:

- I will try to reorganize my handwritten notes, in order to share them with you to provide you with more structured support compared to the simple raw digital file.
- If I don't manage to do it in time, I will anyway share the digital files of the blackboards (OneNote) and, in any case, I strongly invite you to **take your own notes** during the lectures, integrating them with the audio/video recordings that will be made available.

Are there any questions on what was covered today? If everything is perfectly clear, I wish you a good weekend and I will see you on Monday.

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Chapter 3

Lesson 3 -

Chapter 4

Lesson 4 -

Chapter 5

Lesson 5 -